

2 The matrices **A**, **B** and **C** are given by $\mathbf{A} = \begin{pmatrix} 1 & a \\ -1 & 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$, where a is a constant.

(a) By multiplying out the matrices on both sides of the equation, verify that $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$. **[4]**

(b) State the property of matrix multiplication illustrated by this result. **[1]**